



PREMIUM

Full-Length Mock Test

Mock 03

Electronics and Communication Engineering

GATE EC

Total Questions	60
Maximum Marks	108
Duration	180 minutes (3 Hours)
Exam Pattern	MCQ + MSQ + NAT
Negative Marking	MCQ/MSQ: -0.33 per wrong answer
Designed For	EduNeuro Premium

Easy 20%

Moderate 33%

Hard 47%

Based on Previous Year Question Pattern Analysis | Generated by EduNeuro Premium

General Instructions

This question paper contains THREE sections: Section A, Section B, and Section C.

Section A: Multiple Choice Questions (MCQ) — carries one or two marks each. Only ONE option is correct.

Section B: Multiple Select Questions (MSQ) — carries two marks each. ONE OR MORE than one option(s) is/are correct.

Section C: Numerical Answer Type (NAT) — carries one or two marks. Enter the numerical value in the space provided.

Negative Marking: -0.33 marks will be deducted for each wrong MCQ and each wrong MSQ answer.

No negative marking for NAT questions.

There are 60 questions for a total of 100 marks.

Duration: 3 hours (180 minutes).

Use the space provided for rough work. Scribbling on the question paper is permitted.

Do not ask for clarification from the invigilator. Interpret questions as given.

Paper Structure

Section	Type	Questions	Marks/Q	Total Marks	Negative
A	MCQ	35	1 / 2	35	-0.33
B	MSQ	5	2	10	-0.33
C	NAT	20	1 / 2	35	None
Total		60		100	

Marking Scheme

Correct MCQ: +1 mark (or +2 marks as specified per question)

Wrong MCQ: -0.33 marks

Correct MSQ: +2 marks

Wrong MSQ (partial or no match): -0.33 marks

Correct NAT: +1 mark (or +2 marks as specified per question)

No answer / Unattempted: 0 marks

NAT wrong answer: 0 marks (no negative marking)

Section A — Multiple Choice Questions (MCQ)

Q.1 [1 Mark] [EASY] [Network Theory — Thevenin/Norton]

Thevenin equivalent replaces any linear network by:

- A) V-source + Resistor
- B) I-source + Resistor
- C) V-source + Inductor
- D) I-source + Capacitor

Q.2 [1 Mark] [EASY] [Signals & Systems — Convolution]

Impedance of inductor L in s-domain:

- A) $1/s$
- B) sL
- C) $1/(sL)$
- D) s/L

Q.3 [1 Mark] [EASY] [Control Systems — Routh-Hurwitz]

Routh-Hurwitz determines:

- A) Steady-state error
- B) RHP pole count
- C) Damping ratio
- D) Natural frequency

Q.4 [1 Mark] [EASY] [Digital Electronics — ADC/DAC]

BJT in active region:

- A) Both junctions forward
- B) Both reverse
- C) EB forward, CB reverse
- D) EB reverse, CB forward

Q.5 [1 Mark] [EASY] [Analog Electronics — Oscillators]

8085 microprocessor address lines:

- A) 8
- B) 16
- C) 20
- D) 32

Q.6 [1 Mark] [EASY] [EMFT — EM Waves]

FM modulation index, $v = F$, $f = 75\text{kHz}$, $f_m = 15\text{kHz}$:

- A) 3
- B) 5
- C) 10
- D) 15

Q.7 [1 Mark] [EASY] [Communication — Pulse Modulation]

Intrinsic impedance of free space η $\propto f$

A) 50:•

B) 120< ©

C) 377:•

D) Both B and C

Q.8 [1 Mark] [EASY] [Microprocessors — Programming]

A 4-bit DAC with $V_{ref}=5V$ has resolution:

- A) 0.3125V
 - B) 0.5V
 - C) 1V
 - D) 0.25V
-

Q.9 [1 Mark] [EASY] [Network Theory — AC Circuits]

A JK flip-flop with $J=K=1$ toggles on:

- A) Every clock
 - B) Only J change
 - C) Only K change
 - D) Never
-

Q.10 [1 Mark] [EASY] [Signals & Systems — LTI Systems]

Binary 1011 to Gray code:

- A) 1011
 - B) 1110
 - C) 1101
 - D) 0111
-

Q.11 [1 Mark] [EASY] [Control Systems — Bode/Nyquist]

Polarization in EM waves refers to:

- A) Electric field direction
 - B) Magnetic field direction
 - C) Wave propagation direction
 - D) Frequency variation
-

Q.12 [1 Mark] [EASY] [Digital Electronics — Combinational Circuits]

Nyquist rate for signal with max frequency 5kHz:

- A) 5 kHz
 - B) 10 kHz
 - C) 2.5 kHz
 - D) 15 kHz
-

Q.13 [2 Marks] [MODERATE] [Analog Electronics — Oscillators]

Thevenin equivalent replaces any linear network by:

- A) V-source + Resistor
 - B) I-source + Resistor
 - C) V-source + Inductor
 - D) I-source + Capacitor
-

Q.14 [2 Marks] [MODERATE] [EMFT — Waveguides]

Impedance of inductor L in s-domain:

- A) $1/s$

B) SL

C) $1/(SL)$

D) s/L

Q.15 [2 Marks] [MODERATE] [Communication — Digital Modulation]

Routh-Hurwitz determines:

- A) Steady-state error
- B) RHP pole count
- C) Damping ratio
- D) Natural frequency

Q.16 [2 Marks] [MODERATE] [Microprocessors — Interrupts]

BJT in active region:

- A) Both junctions forward
- B) Both reverse
- C) EB forward, CB reverse
- D) EB reverse, CB forward

Q.17 [2 Marks] [MODERATE] [Network Theory — Laplace Circuits]

8085 microprocessor address lines:

- A) 8
- B) 16
- C) 20
- D) 32

Q.18 [2 Marks] [MODERATE] [Signals & Systems — Laplace Transform]

FM modulation index, $v = F$, $f = 75\text{kHz}$, $f_m = 15\text{kHz}$:

- A) 3
- B) 5
- C) 10
- D) 15

Q.19 [2 Marks] [MODERATE] [Control Systems — Compensators]

Intrinsic impedance of free space, $r = \frac{1}{\epsilon_0 \mu_0} \sqrt{\frac{\mu_0}{\epsilon_0}}$

- A) 50 Ω
- B) 120 Ω
- C) 377 Ω
- D) Both B and C

Q.20 [2 Marks] [MODERATE] [Digital Electronics — Memory]

A 4-bit DAC with $V_{ref} = 5\text{V}$ has resolution:

- A) 0.3125V
- B) 0.5V
- C) 1V
- D) 0.25V

Q.21 [2 Marks] [MODERATE] [Analog Electronics — Oscillators]

A JK flip-flop with $J = K = 1$ toggles on:

- A) Every clock

B) Only J change

C) Only K change

D) Never

Q.22 [2 Marks] [MODERATE] [EMFT — Magnetostatics]

Binary 1011 to Gray code:

- A) 1011
- B) 1110
- C) 1101
- D) 0111

Q.23 [2 Marks] [MODERATE] [Communication — Information Theory]

Polarization in EM waves refers to:

- A) Electric field direction
- B) Magnetic field direction
- C) Wave propagation direction
- D) Frequency variation

Q.24 [2 Marks] [MODERATE] [Microprocessors — Programming]

Nyquist rate for signal with max frequency 5kHz:

- A) 5 kHz
- B) 10 kHz
- C) 2.5 kHz
- D) 15 kHz

Q.25 [2 Marks] [MODERATE] [Network Theory — AC Circuits]

Thevenin equivalent replaces any linear network by:

- A) V-source + Resistor
- B) I-source + Resistor
- C) V-source + Inductor
- D) I-source + Capacitor

Q.26 [2 Marks] [MODERATE] [Signals & Systems — Laplace Transform]

Impedance of inductor L in s-domain:

- A) $1/s$
- B) sL
- C) $1/(sL)$
- D) s/L

Q.27 [2 Marks] [MODERATE] [Control Systems — Bode/Nyquist]

Routh-Hurwitz determines:

- A) Steady-state error
- B) RHP pole count
- C) Damping ratio
- D) Natural frequency

Q.28 [2 Marks] [MODERATE] [Digital Electronics — Memory]

BJT in active region:

- A) Both junctions forward

B) Both reverse

C) EB forward, CB reverse

D) EB reverse, CB forward

Q.29 [2 Marks] [MODERATE] [Analog Electronics — Op-Amps]

8085 microprocessor address lines:

- A) 8
- B) 16
- C) 20
- D) 32

Q.30 [2 Marks] [MODERATE] [EMFT — Electrostatics]

FM modulation index ;" $v = F$, " $f = 75\text{kHz}$, $f_m = 15\text{kHz}$:

- A) 3
- B) 5
- C) 10
- D) 15

Q.31 [2 Marks] [MODERATE] [Communication — Information Theory]

Intrinsic impedance of free space ;r $\$f$

- A) 50:•
- B) 120< ©
- C) 377:•
- D) Both B and C

Q.32 [2 Marks] [MODERATE] [Microprocessors — 8085/8086]

A 4-bit DAC with $V_{ref} = 5\text{V}$ has resolution:

- A) 0.3125V
- B) 0.5V
- C) 1V
- D) 0.25V

Q.33 [2 Marks] [HARD] [Network Theory — Two-port Networks]

A JK flip-flop with $J = K = 1$ toggles on:

- A) Every clock
- B) Only J change
- C) Only K change
- D) Never

Q.34 [2 Marks] [HARD] [Signals & Systems — Z-transform]

Binary 1011 to Gray code:

- A) 1011
- B) 1110
- C) 1101
- D) 0111

Q.35 [2 Marks] [HARD] [Control Systems — Compensators]

Polarization in EM waves refers to:

- A) Electric field direction

B) Magnetic field direction

C) Wave propagation direction

D) Frequency variation

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Section B — Multiple Select Questions (MSQ)

Q.36 [2 Marks] [HARD] [Digital Electronics — Sequential Circuits]

Which are closed-loop control components?

- A) Controller
 - B) Comparator
 - C) Plant
 - D) Feedback path
 - E) Reference input
-

Q.37 [2 Marks] [HARD] [Analog Electronics — Feedback]

Which improve steady-state error in control?

- A) Increase K_p
 - B) Add integral
 - C) Add derivative
 - D) Add lag
 - E) Add lead
-

Q.38 [2 Marks] [HARD] [EMFT — Transmission Lines]

Which are digital modulation schemes?

- A) ASK
 - B) FSK
 - C) PSK
 - D) AM
 - E) QAM
-

Q.39 [2 Marks] [HARD] [Communication — AM/FM]

Which are statically indeterminate structures?

- A) Fixed-fixed beam
 - B) Cantilever
 - C) Continuous beam
 - D) Simply supported
 - E) Propped cantilever
-

Q.40 [2 Marks] [HARD] [Microprocessors — Interfacing]

Which are closed-loop control components?

- A) Controller
 - B) Comparator
 - C) Plant
 - D) Feedback path
 - E) Reference input
-

Section C — Numerical Answer Type (NAT)

Q.51 [2 Marks] [HARD] [Control Systems — Block Diagrams]

$G(s) = K/(s(s+2))$. K for marginal stability?

Answer: _____

Q.52 [2 Marks] [HARD] [Digital Electronics — ADC/DAC]

CE amplifier, $R_E = 1k\Omega$, $R_C = 10k\Omega$, $V_{CC} = 10V$, $\beta = 100$. A_v ?

Answer: _____

Q.53 [2 Marks] [HARD] [Analog Electronics — Oscillators]

Parallel plate capacitor $A = 0.1m^2$, $d = 1mm$. C (pF)?

Answer: _____

Q.54 [2 Marks] [HARD] [EMFT — Waveguides]

8085 max memory (KB)?

Answer: _____

Q.55 [2 Marks] [HARD] [Communication — Digital Modulation]

A 3dB bandwidth filter: $|H(j\omega)| = 1/\sqrt{1 + (\omega/\omega_c)^2}$. ω_c at 3dB, $|H(j\omega)| = 1/\sqrt{2}$?

Answer: _____

Q.56 [2 Marks] [HARD] [Microprocessors — Programming]

Digital gate: AND of 4 inputs, each 5V logic. Output (V)?

Answer: _____

Q.57 [2 Marks] [HARD] [Network Theory — Three-phase]

3-phase load: 10A/phase, 0.8pf, 415V. Total real power (W)?

Answer: _____

Q.58 [2 Marks] [HARD] [Signals & Systems — Fourier Series]

For $x(t) = e^{-2t}u(t)$, ROC is $\text{Re}(s) > ?$:

Answer: _____

Q.59 [2 Marks] [HARD] [Control Systems — Compensators]

$G(s) = K/(s(s+2))$. K for marginal stability?

Answer: _____

Q.60 [2 Marks] [HARD] [Digital Electronics — Sequential Circuits]

CE amplifier ;#Ó Â &3ÓV³©, $R_E = 1k$: 'à $|A_v|$ "H ?

Answer: _____

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Answer Key

Verify your answers after completing the full paper.

Q.1:	A	Q.2:	B	Q.3:	B
Q.4:	C	Q.5:	B	Q.6:	B
Q.7:	D	Q.8:	A	Q.9:	A
Q.10:	B	Q.11:	A	Q.12:	B
Q.13:	A	Q.14:	B	Q.15:	B
Q.16:	C	Q.17:	B	Q.18:	B
Q.19:	D	Q.20:	A	Q.21:	A
Q.22:	B	Q.23:	A	Q.24:	B
Q.25:	A	Q.26:	B	Q.27:	B
Q.28:	C	Q.29:	B	Q.30:	B
Q.31:	D	Q.32:	A	Q.33:	A
Q.34:	B	Q.35:	A	Q.36:	ABCDE
Q.37:	ABD	Q.38:	ABCE	Q.39:	ACE
Q.40:	ABCDE	Q.41:	ABD	Q.42:	ABCE
Q.43:	ACE	Q.44:	ABCDE	Q.45:	ABD
Q.46:	ABCE	Q.47:	ACE	Q.48:	ABCDE
Q.49:	ABD	Q.50:	ABCE	Q.51:	4
Q.52:	500	Q.53:	885	Q.54:	64
Q.55:	0.707	Q.56:	5	Q.57:	5744
Q.58:	2	Q.59:	4	Q.60:	500

Detailed Solutions

Question 1 [Network Theory — Thevenin/Norton]

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: V_{th} in series with R_{th} . Norton: I_n in parallel with R_n .

Question 2 [Signals & Systems — Convolution]

Impedance of inductor L in s -domain:

Answer: (B)

$Z_L(s) = sL$ (voltage leads current by 90°).

Question 3 [Control Systems — Routh-Hurwitz]

Routh-Hurwitz determines:

Answer: (B)

Counts poles in Right Half-Plane to assess stability.

Question 4 [Digital Electronics — ADC/DAC]

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 5 [Analog Electronics — Oscillators]

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !' 64KB memory addressing.

Question 6 [EMFT — EM Waves]

FM modulation index ;" $v = F$, " $f = 75\text{kHz}$, $f_m = 15\text{kHz}$:

Answer: (B)

;" $\beta = f/f_m = 75/15 = 5$.

Question 7 [Communication — Pulse Modulation]

Intrinsic impedance of free space ;r η

Answer: (D)

;" $\eta = \sqrt{\mu_0/\epsilon_0} \approx 377 \Omega$

Question 8 [Microprocessors — Programming]

A 4-bit DAC with $V_{ref} = 5V$ has resolution:

Answer: (A)

Resolution = $5V/16 = 0.3125V$.

Question 9 [Network Theory — AC Circuits]

A JK flip-flop with $J=K=1$ toggles on:

Answer:

$J=K=1$ is toggle mode. Output inverts each clock.

Question 10 [Signals & Systems — LTI Systems]

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits !' 1110.

Question 11 [Control Systems — Bode/Nyquist]

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 12 [Digital Electronics — Combinational Circuits]

Nyquist rate for signal with max frequency 5kHz:

Answer: (B)

Nyquist rate = $2 \times f_{\text{max}} = 2 \times 5 = 10$ kHz.

Question 13 [Analog Electronics — Oscillators]

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: V_{th} in series with R_{th} . Norton: I_n in parallel with R_n .

Question 14 [EMFT — Waveguides]

Impedance of inductor L in s -domain:

Answer: (B)

$Z_L(s) = sL$ (voltage leads current by 90°).

Question 15 [Communication — Digital Modulation]

Routh-Hurwitz determines:

Answer: (B)

Counts poles in Right Half-Plane to assess stability.

Question 16 [Microprocessors — Interrupts]

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 17 [Network Theory — Laplace Circuits]

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !' 64KB memory addressing.

Question 18 [Signals & Systems — Laplace Transform]

FM modulation index ;" $v \rightarrow F$, " $f=75\text{kHz}$, $f_m=15\text{kHz}$:

Answer:

(B)

$$\therefore \frac{f}{f_m} = 75/15 = 5.$$

Question 19 [Control Systems — Compensators]

Intrinsic impedance of free space Z_0

Answer: (D)

$$Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \approx 377 \Omega.$$

Question 20 [Digital Electronics — Memory]

A 4-bit DAC with $V_{ref}=5V$ has resolution:

Answer: (A)

$$\text{Resolution} = 5V/16 = 0.3125V.$$

Question 21 [Analog Electronics — Oscillators]

A JK flip-flop with $J=K=1$ toggles on:

Answer: (A)

$J=K=1$ is toggle mode. Output inverts each clock.

Question 22 [EMFT — Magnetostatics]

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits $\Rightarrow 1110$.

Question 23 [Communication — Information Theory]

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 24 [Microprocessors — Programming]

Nyquist rate for signal with max frequency 5kHz:

Answer: (B)

$$\text{Nyquist rate} = 2 \times f_{\max} = 2 \times 5 = 10 \text{ kHz}.$$

Question 25 [Network Theory — AC Circuits]

Thevenin equivalent replaces any linear network by:

Answer: (A)

Thevenin: V_{th} in series with R_{th} . Norton: I_n in parallel with R_n .

Question 26 [Signals & Systems — Laplace Transform]

Impedance of inductor L in s -domain:

Answer: (B)

$$Z_L(s) = sL \text{ (voltage leads current by } 90^\circ \text{)}.$$

Question 27 [Control Systems — Bode/Nyquist]

Routh-Hurwitz determines:

Answer:

(B)

Counts poles in Right Half-Plane to assess stability.

Question 28 [Digital Electronics — Memory]

BJT in active region:

Answer: (C)

Active: Emitter-Base forward, Collector-Base reverse biased.

Question 29 [Analog Electronics — Op-Amps]

8085 microprocessor address lines:

Answer: (B)

8085 has 16-bit address bus !' 64KB memory addressing.

Question 30 [EMFT — Electrostatics]

FM modulation index ;" $v = F$, $f = 75\text{kHz}$, $f_m = 15\text{kHz}$:

Answer: (B)

;" $\Rightarrow f/f_m = 75/15 = 5$.

Question 31 [Communication — Information Theory]

Intrinsic impedance of free space ;" μ_0/ϵ_0

Answer: (D)

;" $\Rightarrow \mu_0/\epsilon_0 = 120\pi \approx 377 \Omega$.

Question 32 [Microprocessors — 8085/8086]

A 4-bit DAC with $V_{ref} = 5V$ has resolution:

Answer: (A)

Resolution = $5V/16 = 0.3125V$.

Question 33 [Network Theory — Two-port Networks]

A JK flip-flop with $J=K=1$ toggles on:

Answer: (A)

$J=K=1$ is toggle mode. Output inverts each clock.

Question 34 [Signals & Systems — Z-transform]

Binary 1011 to Gray code:

Answer: (B)

Gray: same MSB, XOR of consecutive bits !' 1110.

Question 35 [Control Systems — Compensators]

Polarization in EM waves refers to:

Answer: (A)

Polarization = direction of the electric field vector.

Question 36 [Digital Electronics — Sequential Circuits]

Which are closed-loop control components?

Answer:

(A, B, C, D, E)

All listed are components of a feedback control loop.

Question 37 [Analog Electronics — Feedback]

Which improve steady-state error in control?

Answer: (A, B, D)

K_p, integral action, lag compensator reduce SSE.

Question 38 [EMFT — Transmission Lines]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 39 [Communication — AM/FM]

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 40 [Microprocessors — Interfacing]

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 41 [Network Theory — Laplace Circuits]

Which improve steady-state error in control?

Answer: (A, B, D)

K_p, integral action, lag compensator reduce SSE.

Question 42 [Signals & Systems — LTI Systems]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 43 [Control Systems — Routh-Hurwitz]

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 44 [Digital Electronics — ADC/DAC]

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 45 [Analog Electronics — BJT Amplifiers]

Which improve steady-state error in control?

Answer:

(A, B, D)

K_p, integral action, lag compensator reduce SSE.

Question 46 [EMFT — Waveguides]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 47 [Communication — Pulse Modulation]

Which are statically indeterminate structures?

Answer: (A, C, E)

Fixed-fixed, continuous, propped cantilever have redundancy.

Question 48 [Microprocessors — Interrupts]

Which are closed-loop control components?

Answer: (A, B, C, D, E)

All listed are components of a feedback control loop.

Question 49 [Network Theory — Three-phase]

Which improve steady-state error in control?

Answer: (A, B, D)

K_p, integral action, lag compensator reduce SSE.

Question 50 [Signals & Systems — Convolution]

Which are digital modulation schemes?

Answer: (A, B, C, E)

ASK, FSK, PSK, QAM are digital. AM is analog.

Question 51 [Control Systems — Block Diagrams]

$G(s) = K/(s(s+2))$. K for marginal stability?

Answer: (4)

$s^2 + 2s + K = 0$. Marginal stability when $K = 2 \times 2 = 4$.

Question 52 [Digital Electronics — ADC/DAC]

CE amplifier; $V_{CC} = 3V$, $R_E = 1k\Omega$, $|A_v|$?

Answer: (500)

$A_v = -\beta R_E / (r_{be} + \beta R_E) \approx -\beta R_E / r_{be} \approx -500$.

Question 53 [Analog Electronics — Oscillators]

Parallel plate capacitor $A=0.1m^2$, $d=1mm$. C (pF)?

Answer: (885)

$C = \epsilon_0 \epsilon_r A / d = 8.85 \times 10^{-12} \times 1 \times 0.1 / 0.001 = 885 \text{ pF}$.

Question 54 [EMFT — Waveguides]

8085 max memory (KB)?

Answer:

16-bit address bus !' $2^{16} = 65536$ bytes = 64 KB.

Question 55 [Communication — Digital Modulation]

A 3dB bandwidth filter: $|H(j\omega)| = 1$ at $\omega = 0$, $|H(j\omega)| = 0.707$ at $\omega = \omega_c$.

Answer: (0.707)

3dB point = $1/\sqrt{2}$ of maximum response.

Question 56 [Microprocessors — Programming]

Digital gate: AND of 4 inputs, each 5V logic. Output (V)?

Answer: (5)

AND gate outputs HIGH (5V) when all inputs are HIGH.

Question 57 [Network Theory — Three-phase]

3-phase load: 10A/phase, 0.8pf, 415V. Total real power (W)?

Answer: (5744)

$P = 3 \times V_L \times I_L \times \cos\phi = 3 \times 415 \times 10 \times 0.8 = 9960$ W.

Question 58 [Signals & Systems — Fourier Series]

For $x(t) = e^{(-2t)}u(t)$, ROC is $\text{Re}(s) > ?$.

Answer: (2)

$\mathcal{L}\{e^{(-at)}u(t)\}$ has ROC: $\text{Re}(s) > a$. Here $a=2$.

Question 59 [Control Systems — Compensators]

$G(s) = K/(s(s+2))$. K for marginal stability?

Answer: (4)

$s^2 + 2s + K = 0$. Marginal stability when $K = 2 \times 2 = 4$.

Question 60 [Digital Electronics — Sequential Circuits]

CE amplifier; $V_{CC} = 3V$, $R_E = 1k\Omega$, $|A_v| = 500$?

Answer: (500)

$|A_v| = \beta R_E / (r_{be} + R_E) \approx \beta R_E / r_{be} = 500$.

